MUSHABBAR HUSNAIN NOOR

Mechanical Engineer: Aerodynamics & Multiphysics

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LinkedIn | Portfolio

SKILLS & EXPERTISE

Expertise: Aerothermodynamics, Aeroelasticity, Optimization, Avionics Thermal Management, Turbomachinery, Design of Experiment (DoE)
Software: SOLIDWORKS, ANSYS (fluent, pyfluent, CFX, Static Structural, thermal), MATLAB, Python Scripting, COMSOL Multiphysics, Pointwise
Prototyping: Laser Cutting, Wire Cutting, CNC Lathe, CNC Mill, 3D Printing

PROFESSIONAL EXPERIENCE

Aerodynamics Engineer – *BAYKAR Technologies, Pakistan & Turkiye* *February 2024-Present*

- Performed steady state and transient CFD analysis for fixed-wing UAVs, miniature turbojet engine and propeller for on and off-design conditions.
- Analyzed aerodynamic performance metrics, including static and dynamic stability parameters, to ensure flight stability and control.
- Calculating aerodynamics parameters for the development of software in the loop 6-DoF simulator to simulate actual flight dynamics
- Conducted fluid-structure interaction (FSI) based aeroelastic analysis to determine flutter boundaries and establish structural limits.
- Led the evaluation of design modifications, including load-factor assessments on aircraft structures to enhance durability and performance.
- Developed comprehensive VN-diagrams and defined the flight envelope to determine operational safety and performance constraints.
- Performed structural ground testing of aerostructures, validating computational models
- Conducted modal analysis to identify and improve natural frequencies, enhancing structural integrity and mitigating resonance risks.
- Designed air intake for turbojet engine and optimized using adjoint based shape optimization up to 85%.
- Designed and tested engine test frame using aluminum extrusion for adjustable dimensions.
- Assisted in the design whiffle tree apparatus for structural testing of wing and fuselage.
- Participated in aircraft flight testing and validated the CFD model with System ID data.
- Automated the repeated procedures of CFD like domain creation, meshing and solver setting using python scripting and pyfluent.

Aerothermal Research Lead – *Shocks & Stars Engineering, Islamabad* *August 2022-January 2024*

- Performing the aerothermal design analysis of airborne electronics at different flight regimes for AEROFLUX III project
- Conducting analytical and computational analysis to analyze the heat transfer of LRUs mounted in conditioned and unconditioned compartments of high-speed aircraft.
- Performing computational hypersonic aerothermodynamic and combustion analysis of supersonic combustion of airbreathing engine
- Developing novel algorithms and high-fidelity math-models to solve the thermos-fluids problems.
- Applying computational design of experiment-based response surface method for advanced analysis of complete aircraft flight envelope
- Computational and analytical optimization of thermal parameters for optimal heat transfer design points
- To qualify the thermal design of avionics LRUs against altitude and environmental requirements for air worthiness.
- Secured a PKR 1.5 million government research grant.

Aerothermal Lead Engineer – AEROFLUX Lab, Air University January 2022-July 2022

- Performed the analytical and computational thermal airborne avionics system LRU placed in the conditioned bay of high-performance aircraft for AEROFLUX-II project.
- Qualified the thermal design of avionics LRUs against altitude and environmental requirements for air worthiness.
- Thermal analysis of existing design under normal and failure mode for airborne electronics operations
- Designing multiple passive and active heat transfer mode for airborne electronics system
- Secured a PKR 2.5 million government research grant.

Undergraduate Research Assistant – AEROFLUX Lab, Air University June 2021-January 2022

- Conducted thermal analysis of airborne avionics system LRU placed in the conditioned bay of commercial aircraft.
- Performed the analytical and computational thermal analysis of AEROFLUX I project.
- Qualified the thermal design of avionics LRUs under environmental airworthiness.
- Thermal Analysis of existing design under normal and failure mode for LRU operations
- Designing multiple passive and active heat transfer mode
- Secured a PKR 1.5 million government research grant.

PUBLICATIONS

- M. H. Noor, A. Sarosh, U. Riaz, M. Adnan and O. A. A. Awan, "A Figure of Merit Based Computational Heat Transfer Method for Aerothermal Analysis of Airborne Electronics System LRU," https://doi.org/10.1109/IBCAST54850.2022.9990189
- Noor, M.H., Sarosh, A., Khan, A.A. (2024). A Minimal Input Dimensionless Empirical-Corrector (MIDEC) Method for Aerothermal Analysis of Airborne Electronics. Proceedings of the First International Conference on Aeronautical Sciences, Engineering and Technology. ICASET 2023. Springer, Singapore. https://doi.org/10.1007/978-981-99-7775-8_20

EDUCATION

BSc Mechanical Engineering, Air University, Islamabad September 2017-January 2022

AWARDS & CERTIFICATES

- Certified Aerodynamics Engineer, Baykar Technologies
- Judge, CAD Mania Competition, Air University Islamabad
- Pakistan Engineering Council (PEC) Certified Engineer
- Winner, Air Cad Competition, Airtech'19
- PEEF Scholarship (2017-2021)

LANGUAGES

- English (Fluent)
- Urdu (Native)